

Herbivory, predation, disease

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Learning objectives

1. Understand the role of natural enemies for maintaining diversity
2. Explain the relationship between growth and defense and know common defense strategies

last lecture: maintaining diversity, density dependence lecture mentioned natural enemies

Natural enemies

- high density - more susceptible to disease, attract herbivores and predators
- differentiation leads to diversity *species with different natural enemies coexist because more self-limited than limited by other species*
- Janzen-Connell hypothesis: host-specific enemies reduce seedling survival in dense areas
- apparency theory: aggregation is more easily detected by enemies, vulnerable to herbivores
- enemy release hypothesis: invasive plants successful because enemies aren't present

Plant defense against natural enemies

life history theory - talked about tradeoffs plants make between growth, stasis, reproduction because of limited energy

same with defense, allocate energy to maintenance functions, growth or defense

- growth-defense tradeoff
 - grow fast: rapid germination, low defense, higher herbivory *or predation or disease*
 - well-protected: slow growth, high defense, lower herbivory

trading off investment of energy

- dispersal-defense tradeoff
 - diffuse seed dispersal: hard to locate, fewer defenses
 - seed defense: aggregated seeds near parent, more defenses
- attract animals *for dispersal*, deter predators *in species that require animal seed dispersers*

Defense strategies

- physical: leaf toughness, spines, latex, hard seeds
- chemical: secondary metabolites *bitter, spicy, etc*, delayed ripening, masting *local satiation*

Modeling natural enemies

consumer resource model from last time modeled consumer and resource dynamics, similar idea with modeling herbivory

- coupled plant-herbivore dynamics
- a = attack rate *rate that a single herbivore finds and consumes plants*

- e = conversion efficiency *rate that consumed plants converted into new herbivores*
- H = herbivore abundance
- m = herbivore mortality
- N = plant abundance
- K = carrying capacity
- r = intrinsic growth rate

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right) - aNH$$

$$\frac{dH}{dt} = e(aNH) - mH$$